

Fundamentals of Statistical Learning

Definitions cheatsheet

Definition 3.2: Random variable

A *random variable* is a measurable map

$$X : \Omega \longrightarrow \mathbb{R}.$$

It assigns one real number $X(\omega)$ to each outcome ω .

Definition 3.4: Preimage

For a random variable $X : \Omega \rightarrow \mathbb{R}$ and a set $E \subseteq \mathbb{R}$, the *preimage* (or *inverse image*) of E is

$$X^{-1}(E) := \{\omega \in \Omega : X(\omega) \in E\} \subseteq \Omega.$$

To find this event, we collect the outcomes whose outputs fall in E . Whenever $X^{-1}(E)$ is an event, we define

$$\begin{aligned} P(X \in E) &:= P(X^{-1}(E)) \\ &= P(\{\omega \in \Omega : X(\omega) \in E\}). \end{aligned}$$

In particular, for a single value x ,

$$P(X = x) := P(X^{-1}(\{x\})).$$

Here X is the random variable, while x is a particular real number.

Definition 3.5: Distribution of a random variable

The *distribution* of X is the rule that tells us with what probability X takes values in a given set E .

$$P_X(E) = P(X \in E).$$

Definition 3.6: Discrete random variable and PMF

A random variable X is *discrete* if there is a finite or countably infinite set^{*} $\mathcal{X} = \{x_1, x_2, \dots\}$ such that $P(X \in \mathcal{X}) = 1$.

Its *probability mass function* (PMF) is

$$p_X(x) := P(X = x), \quad x \in \mathbb{R}.$$

1. $p_X(x) \geq 0$ for every $x \in \mathbb{R}$.
2. The *support*, in the discrete sense used here, is $\text{supp}(X) := \{x \in \mathbb{R} : p_X(x) > 0\}$. Thus the support lists the values that have positive probability; values outside it have probability zero.
3. $\sum_{x \in \mathcal{X}} p_X(x) = 1$.

Definition 3.7: Continuous random variable and PDF

A random variable X is *continuous* (in the sense used here) if there exists a function f_X , called its *probability density function* (PDF), such that, for every $a \leq b$,

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx.$$

1. $f_X(x) \geq 0$ for every $x \in \mathbb{R}$.
2. Following the convention in the slides, the *support* is $\text{supp}(X) := \{x \in \mathbb{R} : f_X(x) > 0\}$.
3. $\int_{\mathbb{R}} f_X(x) dx = 1$.

Definition 3.8: Cumulative distribution function

Given a random variable X , its *cumulative distribution function* (CDF) is the function $F_X : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F_X(x) := P(X \leq x) = P(X \in (-\infty, x]), \quad x \in \mathbb{R}.$$

It gives the probability accumulated up to and including the threshold x . It is defined for both [discrete](#) and [continuous](#) random variables.

Definition 3.10: Quantile function: general case

Let X be any random variable, discrete or continuous, with CDF F_X . For a probability level $0 < p < 1$, its *quantile function* is

$$Q(p) := \inf\{x \in \mathbb{R} : F_X(x) \geq p\}.$$

It gives the smallest threshold x such that $P(X \leq x)$ is at least p . We also write q_p instead of $Q(p)$.

^{*}A finite set has finitely many elements. A set A is countably infinite if there exists a bijection $f : \mathbb{N} \rightarrow A$.